

Glashow - Weinberg - Salam model

$$SU(2)_{\text{weak}} \times U(1)_{\text{EM}}$$

$$\begin{pmatrix} Z^0 \\ A \end{pmatrix} = \begin{pmatrix} \cos\theta_w & -\sin\theta_w \\ \sin\theta_w & \cos\theta_w \end{pmatrix} \begin{pmatrix} W_3 \\ B \end{pmatrix}$$

$$W_1, W_2 \rightarrow W^\pm$$

θ_w : weak mixing angle

$$g_e = \sqrt{\alpha} = \frac{e}{\sqrt{4\pi}} \quad \text{EM coupling constant}$$

$$g_w = \frac{g_e}{\sin\theta_w} \quad g_z = \frac{g_e}{\sin\theta_w \cos\theta_w}$$

f	c_V	c_A
ν_e, ν_μ, ν_τ	$\frac{1}{2}$	$\frac{1}{2}$
e^-, μ^-, τ^-	$-\frac{1}{2} + 2\sin^2\theta_w$	$-\frac{1}{2}$
u, c, t	$\frac{1}{2} - \frac{4}{3}\sin^2\theta_w$	$\frac{1}{2}$
d, s, b	$-\frac{1}{2} + \frac{2}{3}\sin^2\theta_w$	$-\frac{1}{2}$

Z^0 couplg: $\gamma^\mu (c_V - c_A \gamma^5)$
neutral current

$$M_W = M_Z \cos\theta_w$$

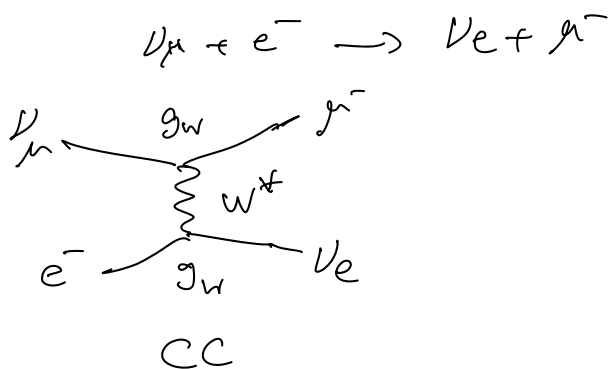
exp. meas. of $M_W, M_Z \Rightarrow$ ~~find~~ provide meas. of θ_w

To verify the theory 1) measure rates $\propto |g_w|^2, |g_z|^2$
 $\Rightarrow$ measure c_V, c_A, θ_w

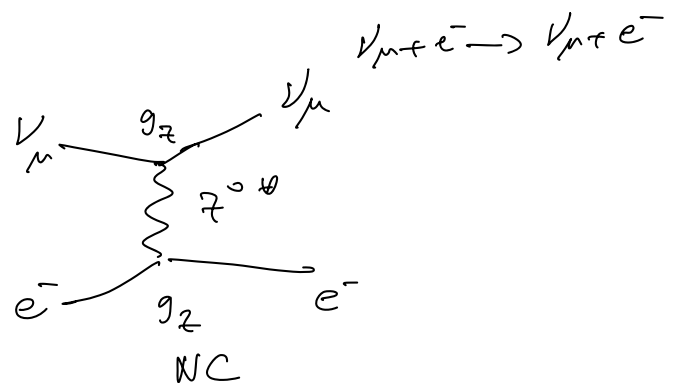
2) produce W, Z , measure M_W, M_Z

$$\theta_w \approx 29^\circ \quad \sin\theta_w \approx 0.23$$

1) measure c_V, c_A (neutral current)



$$M_{CC} \propto g_w^2$$



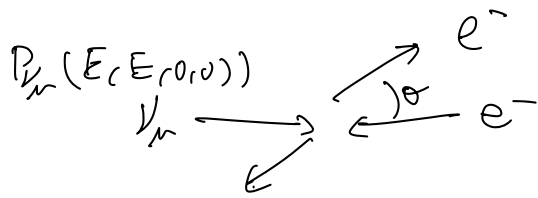
$$M_{NC} \propto g_z^2$$

$$\sigma_{CC}(\nu_\mu + e^- \rightarrow \nu_e + \mu^-) = \sigma(g_w) = \sigma(g_e, \sin\theta_w)$$

$$\sigma_{NC}(\nu_\mu + e^- \rightarrow \nu_\mu + e^-) = \sigma(g_z) = \sigma(g_e, \sin\theta_w, \cos\theta_w)$$

$$q^2 \ll M_Z^2, M_W^2$$

Center of mass reference



measure experimentally
angle of outgoing
lepton in final state

$$\sigma_{NC} \sim \left(\frac{g_1^2}{m_Z^2} \right)^2 E^2 (C_V^2 + C_A^2 + C_V C_A) \quad \text{Griffiths 9.6}$$

$$\frac{\sigma_{NC}}{\sigma_{CC}} = \frac{\#(\nu_\mu + e^- \rightarrow \nu_\mu + e^-)}{\#(\nu_\mu + e^- \rightarrow \nu_e + \mu^-)} = \frac{1}{4} - \sin^2 \theta_W + \frac{4}{3} \sin^2 \theta_W$$

Controlle

theory prediction 0.09

Exp. measu $\approx$ 0.11

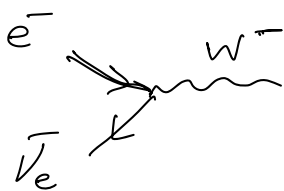
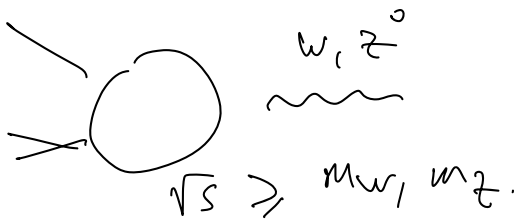
2) production of W, Z on shell. (Direct evidence)

$$G_F = 2.16 \times 10^{-5} \text{ GeV}^{-2} \quad \text{from } n \rightarrow p e^- \bar{\nu}_e \quad \beta \text{ decay in nuclear physics.}$$



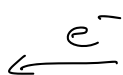
$$\mu \sim \frac{g_W^2}{q^2 - m_W^2}$$

$$q^2 \ll m_W^2 \Rightarrow G_F \approx \frac{g_W^2}{m_W^2} \Rightarrow m_W \approx 80 \text{ GeV}$$

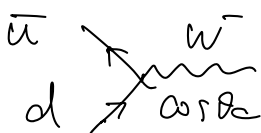


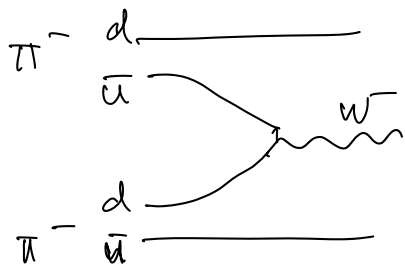
$$E_e \approx \frac{m_W}{2} \Rightarrow E_e \approx 40 \text{ GeV}$$

$$E_{\nu} \approx 40 \text{ GeV}$$



exp. not feasible





$$\pi^- + \pi^- \rightarrow W^- + X$$

$$P_\pi = 40 \text{ GeV} \quad \beta\gamma = \frac{p}{m} = \frac{40}{0.14}$$

$$\approx 400.$$

$$\langle \ell \rangle = \beta\gamma c\tau$$

$$\tau = 10^{-8} \text{ s}$$

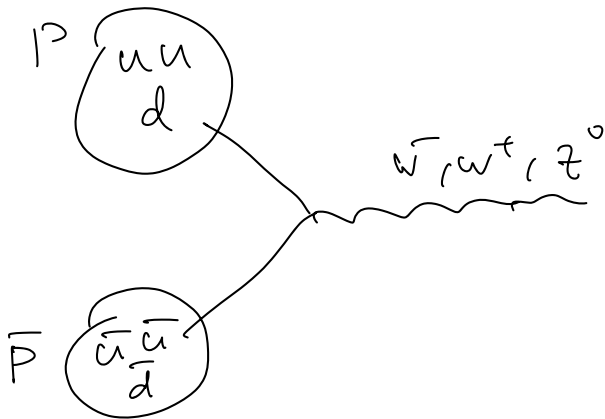
1) Same sign particles to accelerate $\Rightarrow$ two accelerators in parallel.

2) π unstable.

$\Rightarrow$ protons

$$p + p \rightarrow p + p + p + \bar{p} \quad \text{produce } \bar{p}$$

$$p + \bar{p} \rightarrow W + X.$$



At CERN's SPS: proton beam + fixed target

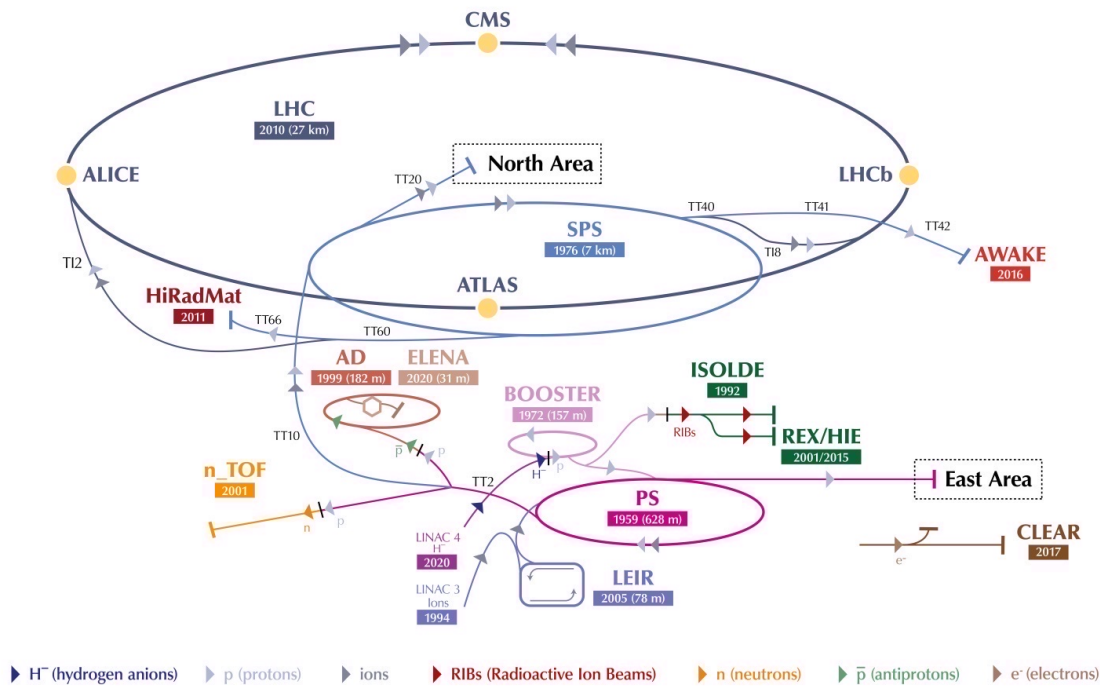
$\bar{p}$ and p have opposite charge

$\Rightarrow$ same magnets can be used.

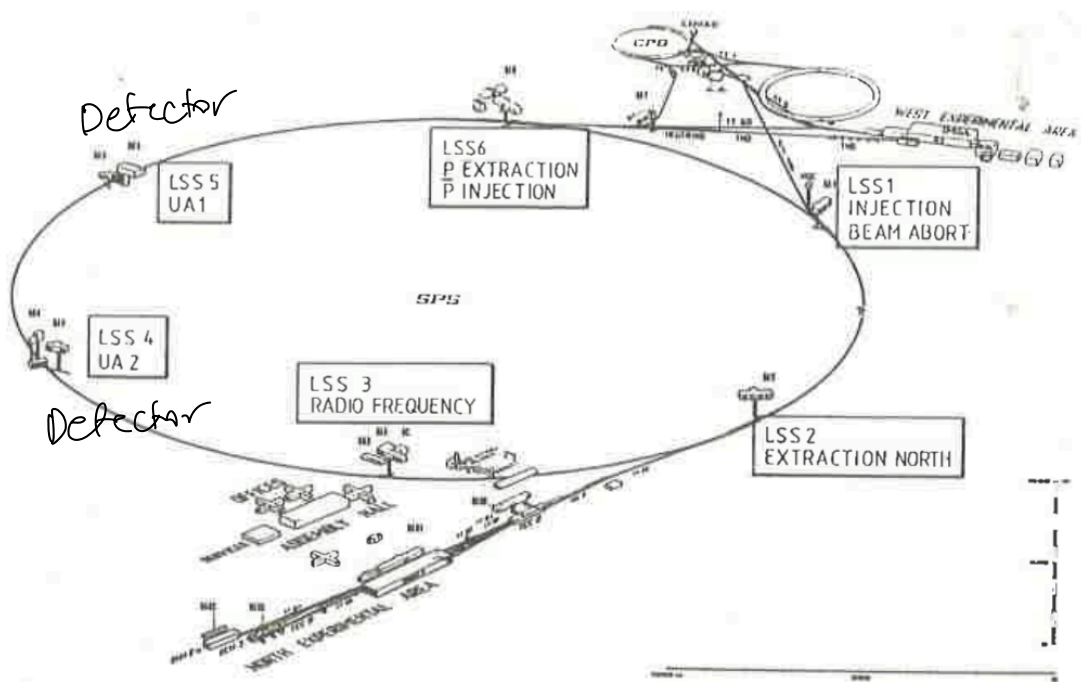
1976: Rubbia proposed to convert SPS $\rightarrow$ Sp $\bar{p}$ S

The CERN accelerator complex

Complexe des accélérateurs du CERN



LHC - Large Hadron Collider // SPS - Super Proton Synchrotron // PS - Proton Synchrotron // AD - Antiproton Decelerator // CLEAR - CERN Linear Electron Accelerator for Research // AWAKE - Advanced WAKEfield Experiment // ISOLDE - Isotope Separator OnLine // REX/HIE - Radioactive Experiment/High Intensity and Energy ISOLDE // LEIR - Low Energy Ion Ring // LINAC - LINear ACcelerator // n_TOF - Neutrons Time Of Flight // HiRadMat - High-Radiation to Materials

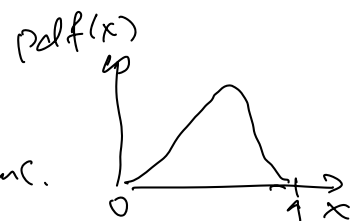


$$\begin{array}{ccc} \xrightarrow{p} & \xleftarrow{\bar{p}} & E \gg m_p. \\ \underline{p}_1 = (E_1, E_1, 0, 0) & \underline{p}_1 = (E_2, -E_2, 0, 0) & \text{momenta of } p, \bar{p} \end{array}$$

$$\underline{p}_u = x_u \underline{p}$$

$$\underline{p}_d = x_d \underline{p}$$

x_u, x_d depend on parton density functions.



$$\langle x \rangle \approx \frac{1}{3}$$

$$\sqrt{s_{ud}} = \sqrt{(E_u + E_d)^2} \approx \langle x \rangle 2 E_p$$

$$= \langle x \rangle \sqrt{s_{pp}}$$

$$SP\bar{P}S: \sqrt{s_{pp}} = 540 \text{ GeV}$$

$$E_p = 270 \text{ GeV} \Rightarrow \text{In average } \sqrt{s_{ud}} \approx \frac{1}{3} 540 =$$

$$\Rightarrow \text{sufficient to produce } u, \bar{u} \text{ on shell.} = 180 \text{ GeV}$$

$$u \longrightarrow \longleftarrow \bar{u}$$

$$u \longrightarrow \longleftarrow \bar{u}$$

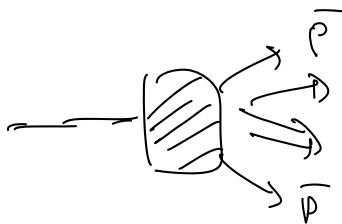
$$u \longrightarrow \longleftarrow \bar{u}$$

long. Lorentz boost changes event by event

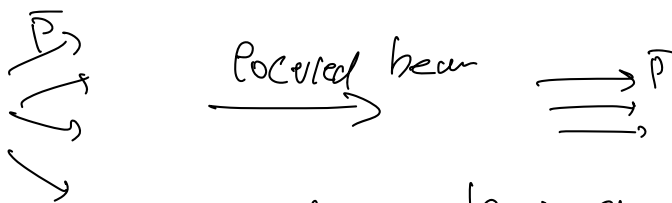
$\Rightarrow$ only rely of conservation of transverse momentum.

$$\text{final state} \sum_i \vec{p}_{T,i} = 0$$

$$p+p \rightarrow p+p+p+\bar{p}$$



thermalization of $\bar{p}$ / stochastic cooling of $\bar{p}$.



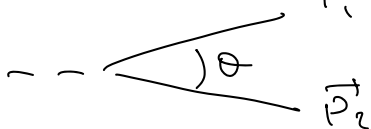
reduce transverse momentum of $\bar{p}$

problem solved by Simon van der Meer

$$N_{\text{events}} = \sigma \cdot \mathcal{L} \cdot \Delta T$$

$$\mathcal{L} \propto \frac{N_p N_{\bar{p}}}{S} \neq \frac{N_p N_{\bar{p}}}{S}$$

$$X \rightarrow 1 + 2$$



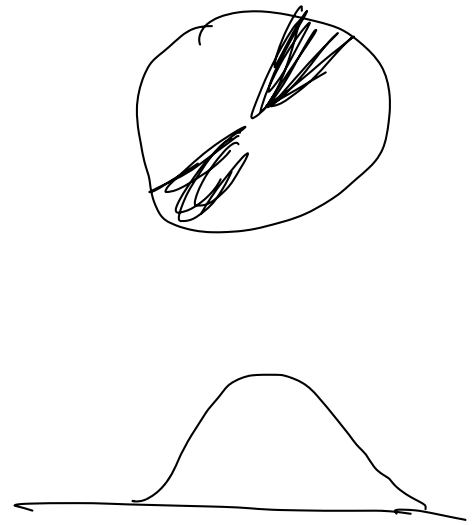
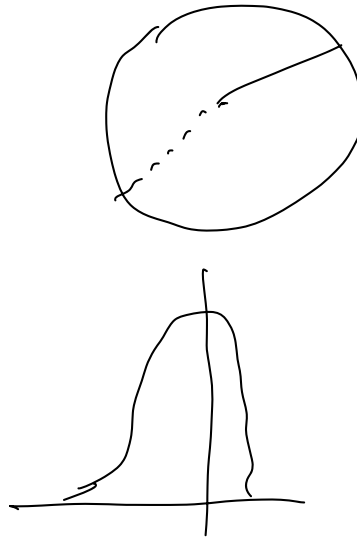
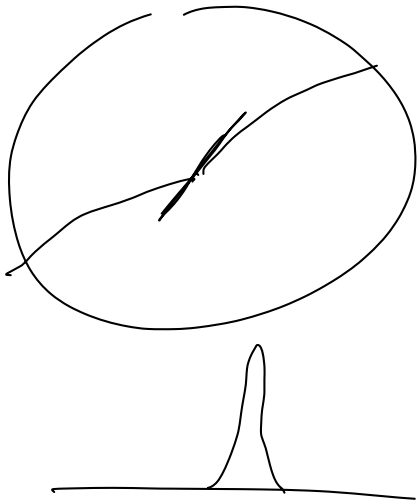
$$M_{12} = \sqrt{(\underline{p}_1 + \underline{p}_2)^2}$$

$$= \sqrt{2 E_1 E_2 (1 - \cos \theta_{12})}$$

$p\bar{p} \rightarrow W, Z + X(\text{hadrons})$
 $\bar{u}^{\pm}, \bar{d}^{\pm}, \bar{s}, \bar{c}, \bar{b}$
 $\rightarrow \ell^+ \ell^-, \nu_e \bar{\nu}_e, \tau^+ \tau^-$
 $\rightarrow \ell^+ \bar{\nu}_e, \ell^+ \nu_e, \tau^+ \bar{\nu}_j$

ij: u, d, s, c, b

Final States: $\ell^+ \ell^-$, $\ell + \text{missing transverse momentum}$, $\text{jet} + \text{jet}$
 Z^0 $W^{\pm}$ Z^0
 W



the **Central Detector**,
a big drift chamber to
track charged particles
in the 0.7 T field of the
dipole magnet

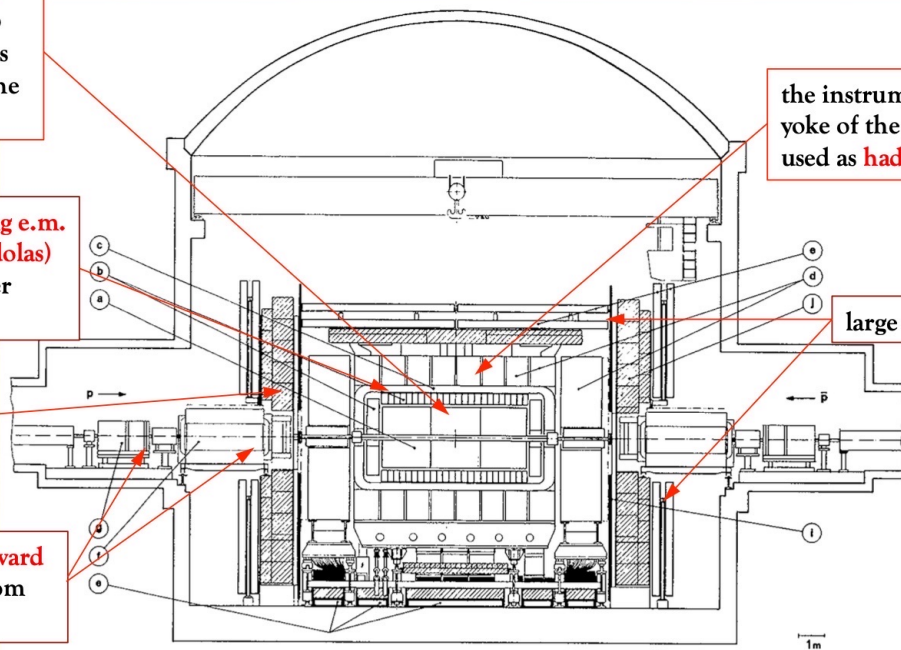
the fine grain **sampling e.m.
calorimeter (the Gondolas)**
readout by wave shifter
bars (BBQ)

end cap detectors
(the bouchons)

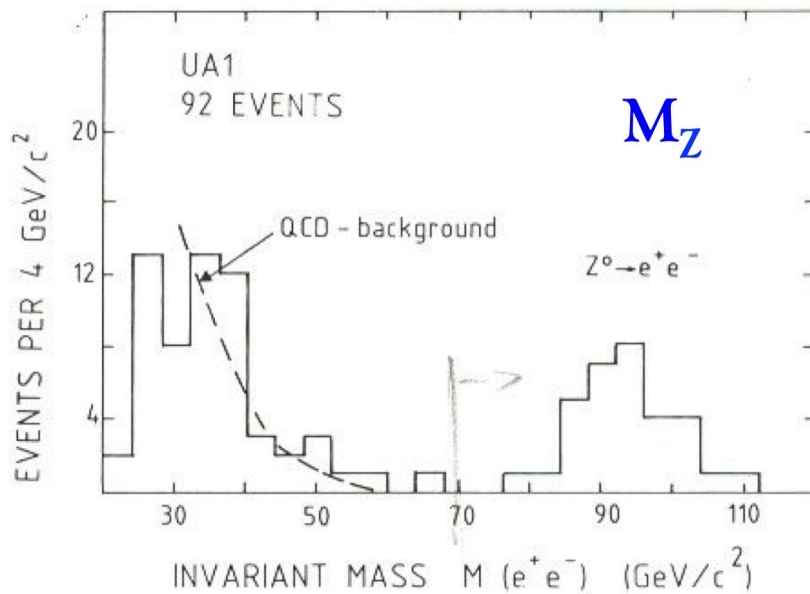
**forward and very forward
detectors** (up to 1° from
the beams)

the instrumented return
yoke of the dipole magnet
used as **hadron calorimeter**

large muon chambers



7



$\sigma_{had} \sim \frac{1}{s}$ # combination

M_{ij}

MERLIN-UA1 VERS. 609

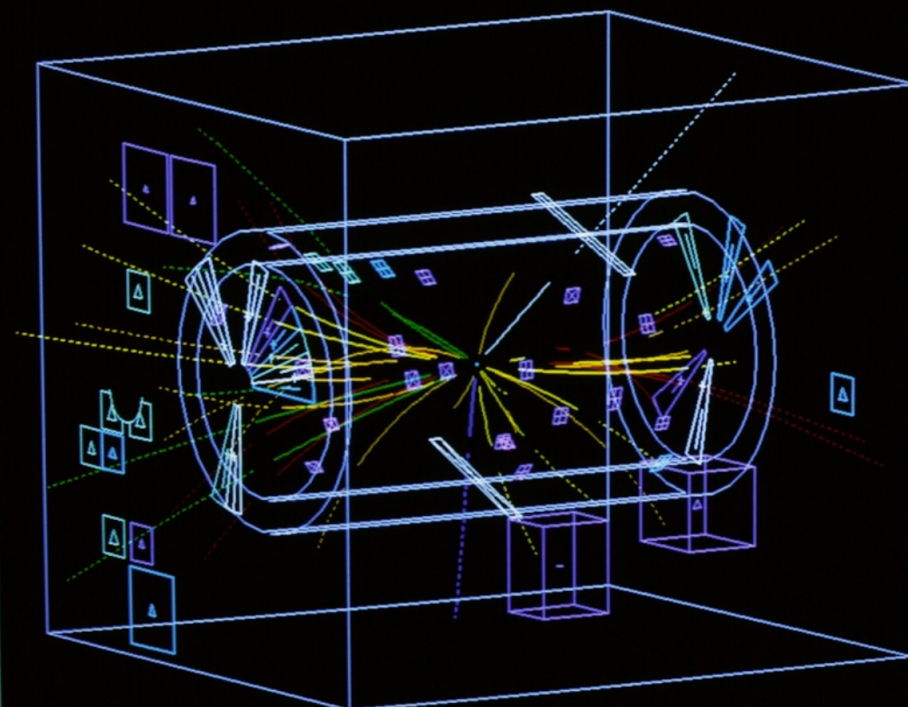
UA1 VERY PRELIMINARY

RUN NO= 6059. EVT NO 1010.

CAMAC DATE 30- 4-85

CAMAC TIME 18:53:

$Z^0 \rightarrow e^+e^-$



+

MERLIN-UA1 VERS. 609

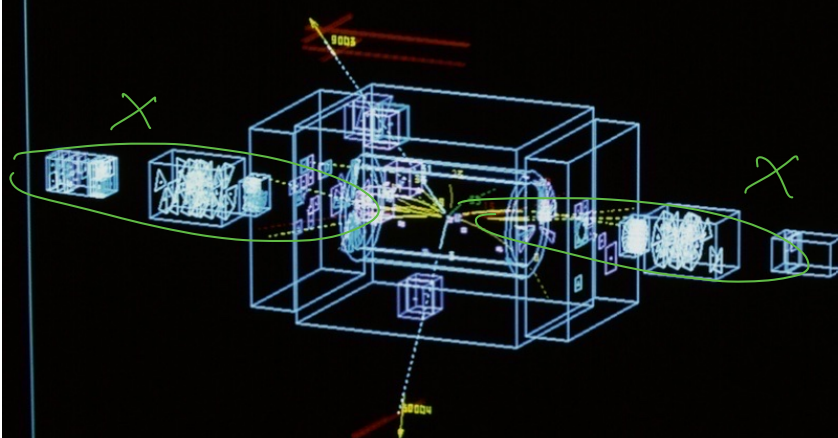
UA1 VERY PRELIMINARY

RUN NO= 8523. EVT NO 831.

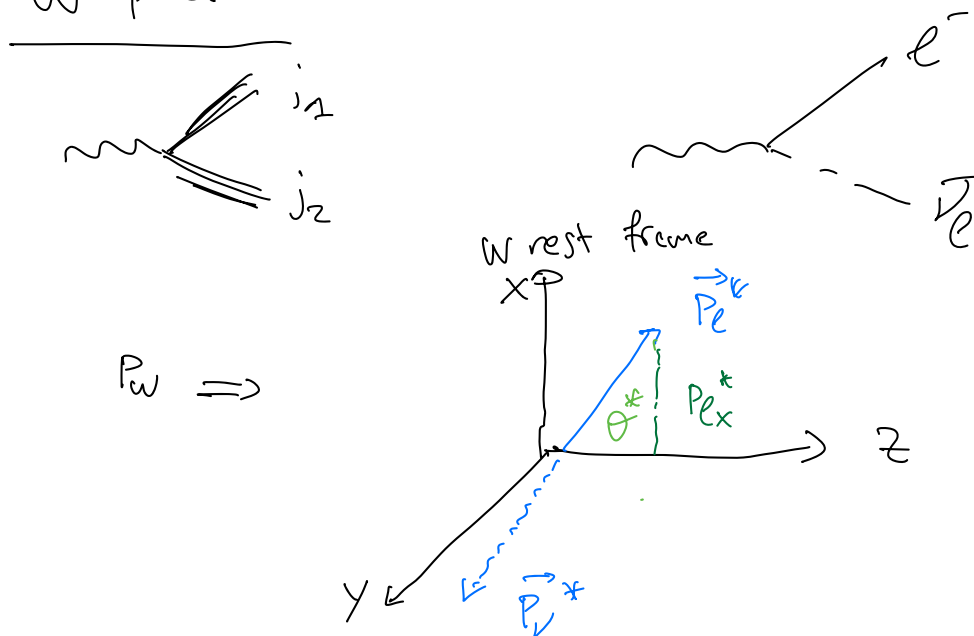
CAMAC DATE 30- 6-85

CAMAC TIME 21:54:

$Z^0 \rightarrow \mu^+\mu^-$



W peak



rest frame

$$e^- \leftarrow W^- \rightarrow \nu_e$$

$$E_e^* = E_{\nu}^* = \frac{m_W}{2}$$

$\Downarrow$

$$p_{\nu}^* = \frac{m_W}{2}$$

$$E_e^* = p_e^* = \frac{m_W}{2}$$

$$p_{e_x}^* = \frac{m_W}{2} \sin \theta^* = p_{e_x}$$

$p_{e_x} \perp$ boost direction.

$$\frac{dn}{dp_{e_x}^*} = \frac{dn}{d\theta^*} \frac{d\theta^*}{dp_{e_x}^*}$$

$$\sin \theta^* = \frac{p_{e_x}}{m_W/2}$$

change of variable.

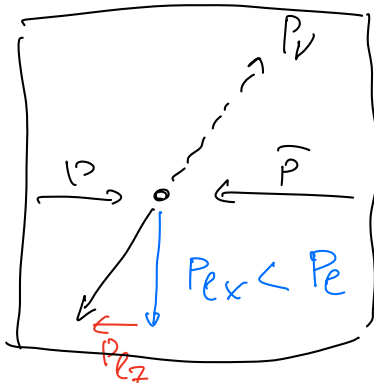
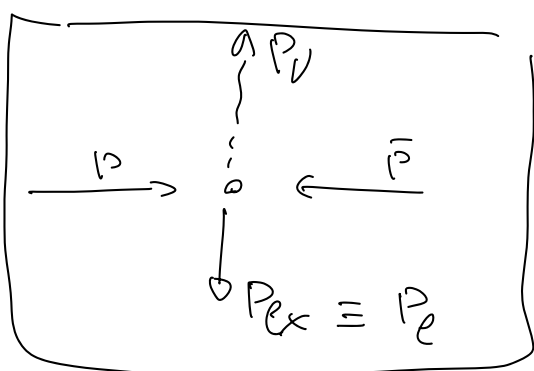
$$\frac{dn}{dp_{e_x}} = \frac{dn}{d\theta^*}$$

$$\frac{1}{\sqrt{\left(\frac{m_W}{2}\right)^2 - (p_{e_x})^2}}$$

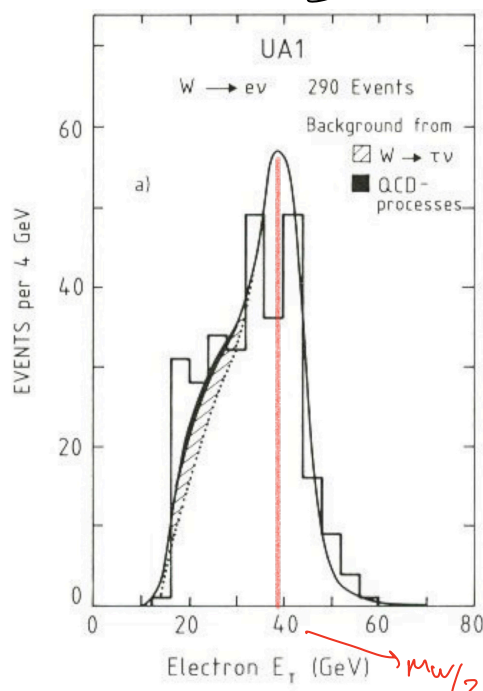
Singularity at $\frac{m_W}{2}$

$$p_{e_x} = \frac{m_W}{2}$$

underlying distribution in W rest frame.



$p_{e_x} < p_e$ because only transverse component



E_T : transverse energy

In alternative:

$$\vec{P}_{\nu_T} = - \sum_{i=1}^{final} \vec{P}_{\tau_i}$$

$$m_W^T = \sqrt{(\vec{P}_{e_T} + \vec{P}_{\nu_T})^2}$$

four-vector invariant mass.

$$m_W^T \leq m_W$$

$$W^- \rightarrow e^- \bar{\nu}_e$$

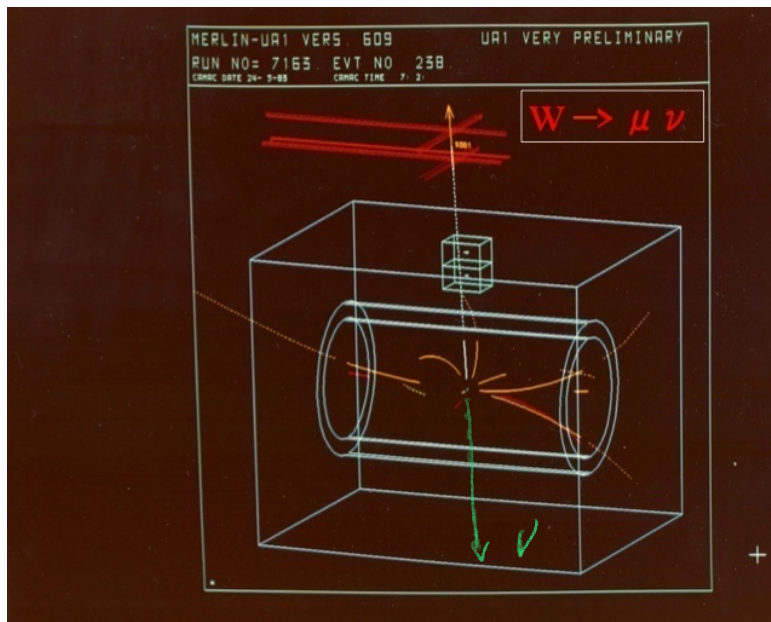
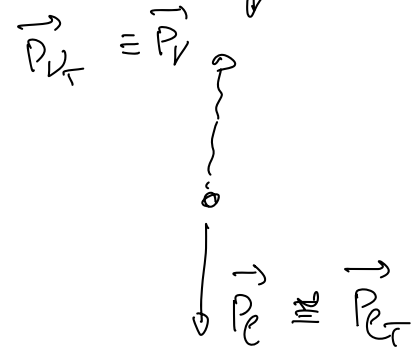
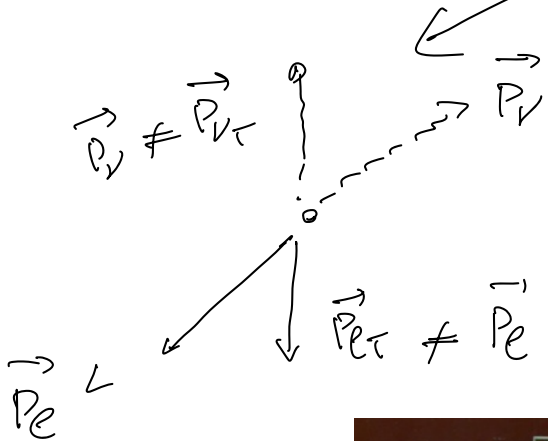
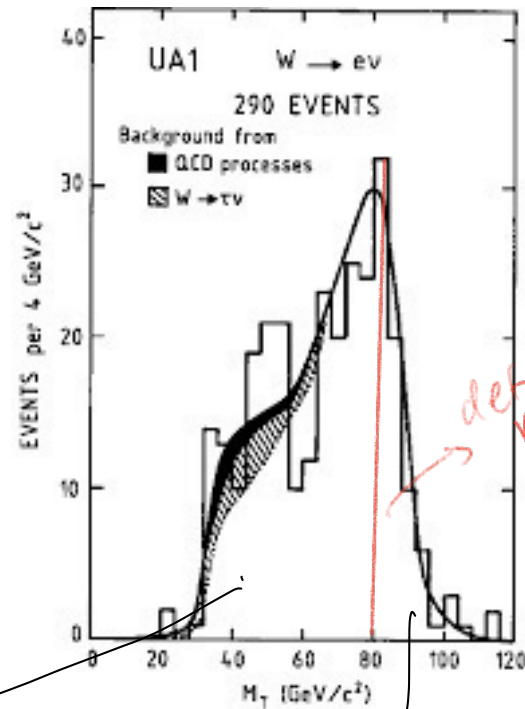
$$W^- \rightarrow \tau^- \bar{\nu}_\tau$$

$$\hookrightarrow e^- \bar{\nu}_e \nu_\tau$$

$$W^- \rightarrow \tau^- \bar{\nu}_\tau$$

$$\hookrightarrow \mu^- \bar{\nu}_\mu \nu_\tau$$

$$\hookrightarrow e^- \bar{\nu}_e \nu_\mu$$



W⁺ DECAY MODES

W⁻ modes are charge conjugates of the modes below.

Mode	Fraction (Γ_i/Γ)	Confidence level
Γ_1 $\ell^+\nu$	[a] (10.86 ± 0.09) %	
Γ_2 $e^+\nu$	(10.71 ± 0.16) %	
Γ_3 $\mu^+\nu$	(10.63 ± 0.15) %	
Γ_4 $\tau^+\nu$	(11.38 ± 0.21) %	
Γ_5 hadrons	(67.41 ± 0.27) %	
Γ_6 $\pi^+\gamma$	< 7 × 10 ⁻⁶	95%
Γ_7 $D_s^+\gamma$	< 1.3 × 10 ⁻³	95%
Γ_8 cX	(33.3 ± 2.6) %	

Z DECAY MODES

Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
Γ_1 e^+e^-	[a] (3.3632 ± 0.0042) %	
Γ_2 $\mu^+\mu^-$	[a] (3.3662 ± 0.0066) %	
Γ_3 $\tau^+\tau^-$	[a] (3.3696 ± 0.0083) %	
Γ_4 $\ell^+\ell^-$	[a,b] (3.3658 ± 0.0023) %	
Γ_5 $\mu^+\mu^-\mu^+\mu^-$		
Γ_6 $\ell^+\ell^-\ell^+\ell^-$	[c] (4.55 ± 0.17) × 10 ⁻⁶	
Γ_7 invisible	[a] (20.000 ± 0.055) %	
Γ_8 hadrons	[a] (69.911 ± 0.056) %	

hadrons: $u\bar{u}, d\bar{d}, s\bar{s}, c\bar{c}, b\bar{b} \times 3$

$\sim \frac{9}{9}$ QCD for hadron. Z

LEP: $\sqrt{s} = m_Z$

$P(Z \rightarrow ee, \mu\mu, \tau\tau + \text{hadrons} + \nu_e\bar{\nu}_e + \nu_\mu\bar{\nu}_\mu + \nu_\tau\bar{\nu}_\tau)$

C, # total # Z⁰ decays.

$\Rightarrow$ measurement of # families.

LEP measure # ν with $m_\nu < \frac{m_Z}{2}$
= 3